UNIT – I

Elastic & Plastic Behaviour – Dislocations

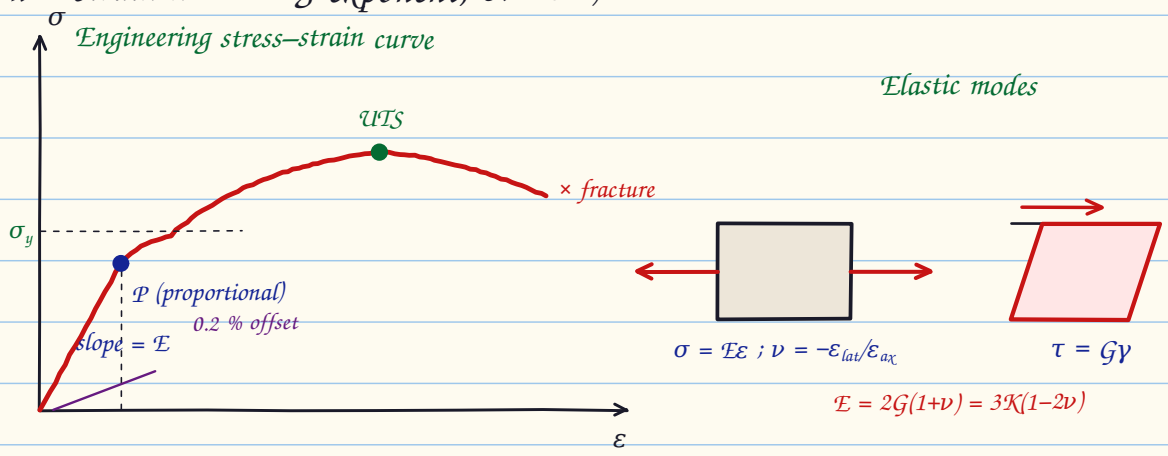
Subject : Mechanical Behaviour of Materials (23A03504c)

Syllabus: Elastic behaviour – Hooke's law ; plastic behaviour – dislocation theory, Burgers vectors, dislocation loops, dislocations in FCC, HCP & BCC lattices, stress fields & energies of dislocations, forces on and between dislocations, slip and twinning.

1. Introduction – Mechanical Behaviour

Mechanical behaviour = response of a material (deformation, yielding, fracture) to applied loads. Engineers need strength, stiffness, ductility, toughness, fatigue & creep resistance.

- **Engineering stress** $\sigma = P/A_0$; **engineering strain** $\varepsilon = \Delta l/l_0$. **True stress** $\sigma_T = P/A = \sigma(1+\varepsilon)$; **true strain** $\varepsilon_T = \ln(l/l_0) = \ln(1+\varepsilon)$ (valid up to necking, constant volume).
- Plastic flow after yield follows $\sigma_T = K\varepsilon_T^{\,n}$ (Hollomon : K = strength coefficient, n = strain-hardening exponent, 0.1–0.5).



- Regions of the stress–strain curve : elastic (reversible, Hooke), yield (onset of plasticity), strain hardening (up to UTS), necking and fracture. Yield strength is usually the **0.2 % offset** stress (line parallel to the elastic slope drawn from $\varepsilon = 0.002$).

- Ductility : % elongation = $(l_f - l_0)/l_0 \times 100$; % reduction of area = $(A_0 - A_f)/A_0 \times 100$. Toughness = area under the σ–ε curve ; resilience $U_r = \sigma_y^2/2E$.

2. Elastic Behaviour – Hooke's Law

Elastic deformation is reversible : the atoms are displaced from their equilibrium positions but return when the load is removed (bond stretching).

$$\sigma = E\,\varepsilon \;\; \tau = G\,\gamma \;\; p = -K(\Delta V/V) \text{ (Hooke's law)}$$

- **E** Young's modulus (tension), **G** shear (rigidity) modulus, **K** bulk modulus. **Poisson's ratio** $\nu = -\varepsilon_{lateral}/\varepsilon_{axial}$ (0.25–0.35 for metals, 0.5 for incompressible rubber, $\nu \le 0.5$).

$$E = 2G(1+\nu) = 3K(1-2\nu) \quad G = E/2(1+\nu) \quad K = E/3(1-2\nu)$$

- **Origin** : interatomic force $F(r)$ – at the equilibrium spacing r_0 the net force is zero ; for small displacement $F \approx S_0(r - r_0)$, where $S_0 = dF/dr$ (bond stiffness). $E \propto S_0/r_0$ ⇒ strong, tightly bonded materials (high T_m) have high E ; E drops slowly with temperature.
- E is a structure-insensitive property (not changed by grain size, cold work, heat treatment) but depends on crystal direction (anisotropy) : Cu $E_{[111]} = 191$ GPa, $E_{[100]} = 67$ GPa.

Material	E (GPa)	G (GPa)	ν	Remark
Steel	200–210	79	0.29	stiff
Cast iron	100–170	45	0.26	
Aluminium	70	26	0.33	1/3 of steel
Copper	120	44	0.34	
Titanium	110	42	0.34	
Glass / concrete	70 / 30	28 / 12	0.22 / 0.2	brittle

Generalised Hooke's law (isotropic, 3-D)

$$\varepsilon_x = (1/E)[\sigma_x - \nu(\sigma_y + \sigma_z)] \quad \gamma_{xy} = \tau_{xy}/G \text{ (and cyclic)}$$

- Dilatation $\Delta = \varepsilon_x + \varepsilon_y + \varepsilon_z = (1-2\nu)(\sigma_x + \sigma_y + \sigma_z)/E = p/K$. Anisotropic crystals need up to 21 independent constants (cubic : C_{11}, C_{12}, C_{44}) ; anisotropy

ratio $A = 2C_{44}/(C_{11} - C_{12})$.

- **Anelasticity** : time-dependent recovery of strain (internal friction) – damping in metals ; **elastic limit** may differ slightly from the proportional limit.

> Ex 1. $E = 200$ GPa, $\nu = 0.30$: $G = 200/2.6 =$ 76.9 GPa ; $K = 200/1.2 =$ 166.7 GPa. A steel rod loaded to $\sigma = 400$ MPa : $\varepsilon = 400/200\,000 = 0.002$; lateral strain $= -0.3(0.002) = -0.0006$. Resilience for $\sigma_y = 500$ MPa : $U_r = 500^2/(2 \times 200\,000) =$ 0.625 MJ/m³.

3. Plastic Behaviour – Why Dislocations?

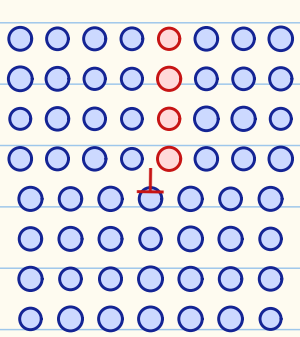
Plastic deformation = permanent shape change by slip (movement of atoms to new equilibrium positions) beyond the elastic limit, at constant volume.

- **Theoretical shear strength** (Frenkel, 1926) : if two perfect planes slide over each other as rigid blocks, $\tau = (G\,b/2\pi\,a)\sin(2\pi x/b) \Rightarrow \tau_{th} \approx G/2\pi \approx G/10 - G/30 \approx$ 3–8 GPa for metals.
- **Observed** yield strength of pure single crystals is only 0.1–10 MPa ($10^{-3} - 10^{-4}$ G) – 1000 times lower ⇒ slip does not occur by simultaneous shift of whole planes but by the motion of line defects, **dislocations** (Orowan, Polanyi, Taylor – 1934).
- Analogy : moving a carpet by a ripple ; caterpillar motion ; only a few bonds are broken at a time ⇒ low stress.

> Ex 2. Cu : $G = 48$ GPa ⇒ $\tau_{th} = 48/(2\pi) = 7.6$ GPa ; measured CRSS ≈ 0.6 MPa ⇒ ratio ≈ 10^4 – evidence for dislocations. Dislocations were first seen directly by TEM (Hirsch, 1956) ; density ρ = line length per unit volume (m^{-2}).

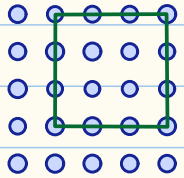
4. Dislocations – Types & Burgers Vector

Edge dislocation

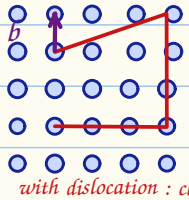


b ⊥ dislocation line ; ξ ⊥ b

Burgers circuit



perfect : closes



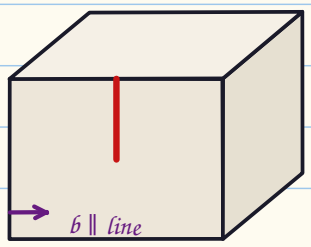
with dislocation : closure failure = b

Edge dislocation

- Formed by an extra half-plane of atoms inserted in a perfect lattice. The dislocation line ξ is the edge of the half-plane. Above the line the lattice is in compression, below in tension. Symbol ⊥ (positive, extra plane above) / ⊤ (negative).
- **Burgers vector b** : the closure failure of a circuit (Burgers circuit) which is closed in the perfect crystal – b is perpendicular to the line in an edge dislocation. Slip occurs on the plane containing b and ξ (a unique glide plane) ; edge can also **climb** (move out of the glide plane by vacancy diffusion – needs thermal activation).
- **Burgers circuit rule** : (FS/RH convention) circuit drawn clockwise looking along ξ ; b = vector from finish to start (F→S).

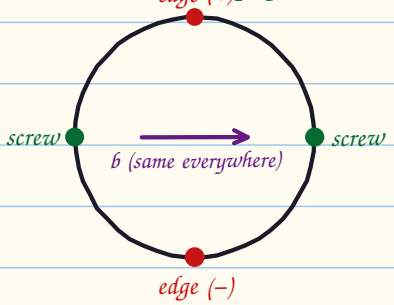
Screw dislocation

Screw dislocation

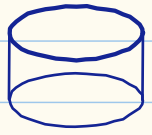


spiral ramp ; shear τ ∥ line

Dislocation loop (glide)



Prismatic loop



b ⊥ loop plane (climb)

vacancy / interstitial disc

- Crystal planes form a continuous spiral ramp (like a screw) around the line ; b is parallel to the line. Produced by shear applied to part of the crystal. Since b ∥ ξ, any plane containing the line is a glide plane ⇒ the screw can **cross-slip** (change

slip plane) – important in recovery and work hardening ; it does not climb.

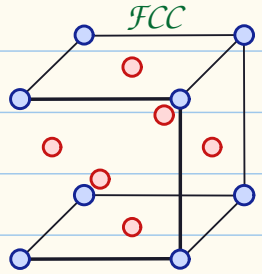
Mixed dislocation & dislocation loops

- A curved dislocation line has edge character where $b \perp \xi$ and screw where $b \parallel \xi$; in between, mixed. The Burgers vector is the same everywhere along the line (a dislocation cannot end inside a crystal – it ends on a surface, boundary, another dislocation (nodes) or closes into a **loop**).
- **Node rule** : at a junction $\Sigma\, b_{in} = \Sigma\, b_{out}$ (like Kirchhoff's current law).
- **Glide loop** : b lies in the plane of the loop – expands/contracts by slip. **Prismatic loop** : $b \perp$ loop plane – formed by condensation of vacancies (vacancy loop) or interstitials (interstitial loop) ; moves only by climb (sessile). **Frank loop** : faulted loop in FCC ($b = a/3\langle 111\rangle$).
- **Dislocation density** ρ : annealed crystal 10^{6}–10^{8} cm^{-2} (10^{10}–10^{12} m^{-2}) ; heavily cold worked 10^{11}–10^{12} cm^{-2} ; ideal (whiskers) 0.

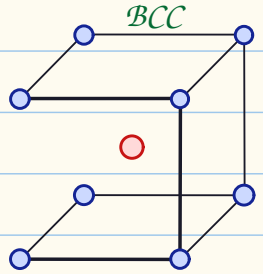
Characteristics summary

Property	Edge	Screw
b vs line ξ	$b \perp \xi$	$b \parallel \xi$
Glide plane	unique ($b \times \xi$)	any plane containing ξ
Direction of motion	$\parallel$ to b	$\perp$ to b (line moves $\perp$ to the shear)
Climb	possible	not possible
Hydrostatic stress field	yes	no (pure shear)
Cross-slip	no	yes

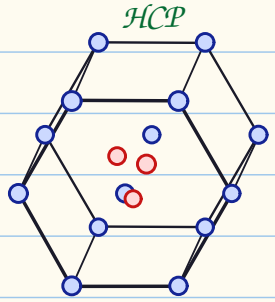
5. Dislocations in FCC, BCC and HCP Lattices



$\{111\}\langle 110\rangle$: 12 systems
$|b| = a/\sqrt{2} = 2.55$ Å (Cu)



$\{110\}\langle 111\rangle$ (+$\{112\},\{123\}$) : 48
$|b| = a\sqrt{3}/2 = 2.48$ Å (Fe)



basal $(0001)\langle 11\text{-}20\rangle$: 3 systems
$|b| = a$ (Zn 2.66 Å)

- **Slip occurs on close-packed planes along close-packed directions** (shortest b ⇒ lowest energy ∝ b^2). Slip system = {plane}⟨direction⟩.

Structure	Slip system	No.	Burgers vector ; length
FCC (Cu, Al, Ni, Au)	{111}⟨110⟩	12	a/2⟨110⟩ ; a/√2
BCC (Fe, W, Cr, Mo)	{110},{112},{123}⟨111⟩	12 + 12 + 24 = 48	a/2⟨111⟩ ; a√3/2
HCP (Zn, Mg, Cd, Ti)	basal (0001)⟨11-20⟩ ; prismatic, pyramidal	3 (basal) + 3 + 6	a/3⟨11-20⟩ ; a (also ⟨c+a⟩)

- **FCC** : 4 {111} planes × 3 ⟨110⟩ directions = 12 systems – very ductile, low CRSS, little temperature dependence. Cu a = 3.615 Å ⇒ $|b|$ = 2.556 Å.
- **BCC** : no truly close-packed plane ; {110} is the densest ; many systems of comparable spacing ⇒ pencil-glide ; CRSS strongly depends on temperature and strain rate (screw dislocations have non-planar cores – high Peierls stress) ⇒ ductile–brittle transition. Fe a = 2.866 Å ⇒ $|b|$ = 2.482 Å.
- **HCP** : only 3 independent basal systems (c/a > 1.63 : Zn, Cd – basal slip, brittle ; c/a ≈ 1.59 Ti, Zr – prismatic slip also ; Mg at room T basal only, non-basal above 225 °C) ⇒ limited ductility, twinning is important.

Partial dislocations & stacking faults (FCC)

- A perfect dislocation in FCC dissociates into two **Shockley partials** joined by a ribbon of stacking fault (…ABCABABC…) :

$a/2[1\,{-1}\,0] \rightarrow a/6[2\,{-1}\,{-1}] + a/6[1\,{-2}\,1]$ (energy lowered since $b_1^2 > b_2^2 + b_3^2$ – Frank's criterion : $a^2/2 > a^2/3$)

- Equilibrium width of the fault (balance of repulsion between partials and fault energy γ) :

$$d \approx G\, b_1 \cdot b_2 / (2\pi\, \gamma_{SF}),\ b_1 \cdot b_2 = a^2/12 \text{ (screw)} ;\ \text{edge : divide by } (1-\nu)$$

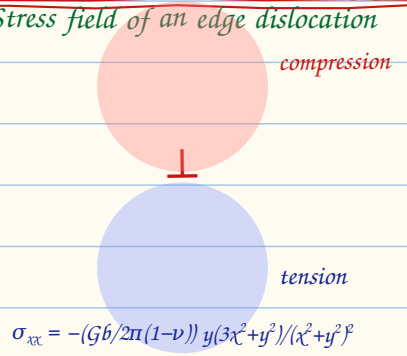
- Stacking-fault energy γ (mJ/m²) : Al 166, Ni 125, Cu 45, Au 32, Ag 22, brass/stainless < 20. High γ (Al) : narrow ribbon, easy cross-slip, easy recovery. Low γ (Cu-alloys, stainless steel) : wide ribbon, difficult cross-slip, rapid work hardening, twinning.

Ex 3. Cu : $a = 3.615$ Å, $G = 48$ GPa, $\gamma = 45$ mJ/m² : $b_1 \cdot b_2 = (3.615\times10^{-10})^2/12 = 1.09\times10^{-20}$ m² ⇒ $d = (48\times10^9 \times 1.09\times10^{-20})/(2\pi \times 0.045) = 1.85$ nm (screw) ; edge $d = 1.85/0.66 = 2.8$ nm (≈ 7–11 atomic spacings).

- **Sessile dislocations** : Lomer–Cottrell lock (formed by reaction of partials on intersecting {111} planes, a/6⟨110⟩ stair-rod) – barrier to glide, strong work hardening in FCC ; Frank sessile loops.
- **Dislocation reactions** : allowed if Σ b conserved and energy (b^2) decreases : e.g. a/2[101] + a/2[01-1]... → a/2[110]. BCC : a/2[111] + a/2[1-1-1] → a[100] (cleavage crack nucleation, Cottrell).

6. Stress Fields & Energy of Dislocations

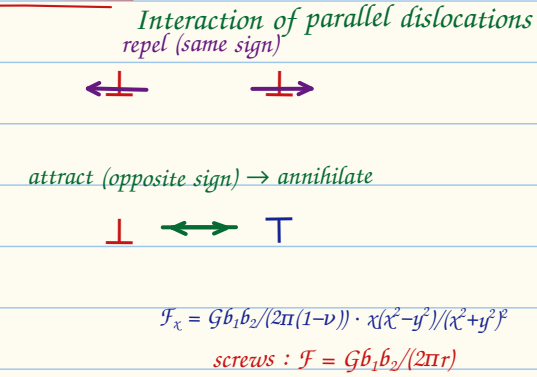
Stress field of an edge dislocation



$\sigma_{xx} = -(Gb/2\pi(1-\nu))\, y(3x^2+y^2)/(x^2+y^2)^2$

$\tau_{xy} = (Gb/2\pi(1-\nu))\, x(x^2-y^2)/(x^2+y^2)^2$

Interaction of parallel dislocations



$F_x = Gb_1b_2/(2\pi(1-\nu)) \cdot x(x^2-y^2)/(x^2+y^2)^2$

screws : $F = Gb_1b_2/(2\pi r)$

Stress field (isotropic elastic continuum)

- **Screw** (cylindrical coordinates, no normal stresses) : $\tau_{\theta z} = G\,b\,/\,(2\pi\,r)$ — varies as 1/r, singular at the core (valid for $r > r_0 \approx b \Rightarrow 2$–$3b$).
- **Edge** (x along b, y normal to the glide plane) :

$$\sigma_{xx} = -\,[Gb/2\pi(1-\nu)]\; y(3x^2+y^2)/(x^2+y^2)^2 \quad \sigma_{yy} = [Gb/2\pi(1-\nu)]\; y(x^2-y^2)/(x^2+y^2)^2$$

$$\tau_{xy} = [Gb/2\pi(1-\nu)]\; x(x^2-y^2)/(x^2+y^2)^2 \quad \sigma_{zz} = \nu(\sigma_{xx} + \sigma_{yy})$$

- Features : compressive σ_{xx} above the glide plane ($y > 0$), tensile below ; τ_{xy} is maximum on the glide plane ($y = 0$) and zero at $x = 0$ and on the lines $x = \pm y$; the stress decays as 1/r – a long-range field, so dislocations interact over large distances.

Strain energy per unit length

- Energy in the elastic field between r_0 (core) and R (outer radius ≈ half the spacing between dislocations ≈ $1/(2\sqrt{\rho})$) :

$$E_{screw} = (G\,b^2/4\pi)\,\ln(R/r_0)\quad E_{edge} = (G\,b^2/4\pi(1-\nu))\,\ln(R/r_0)$$

- Putting $\ln(R/r_0) \approx 4\pi$ ($R \approx 10^3 b$) : $E \approx \alpha\, G\, b^2$, $\alpha = 0.5 - 1$ (J/m) ; **line tension** $T \approx \tfrac{1}{2}\, G\, b^2$ (resists bending – like a stretched rubber string). Edge energy is ≈ 1.5 times that of the screw $(1/(1-\nu))$.
- Energy $\propto b^2 \Rightarrow$ dislocations with the smallest b are most stable ; a dislocation tends to be straight (short) to minimise energy. **Core energy** ≈ $Gb^2/10$ (adds 10–20 %).

Ex 4. Cu : G = 48 GPa, b = 2.55 Å : $Gb^2 = 48\times10^9 \times 6.5\times10^{-20} = 3.12\times10^{-9}$ J/m $\Rightarrow E \approx 0.5\, Gb^2 = 1.56\times10^{-9}$ J/m $\Rightarrow$ per atom length (b) = $1.56\times10^{-9} \times 2.55\times10^{-10} = 4.0\times10^{-19}$ J = 2.5 eV – large compared with kT (0.025 eV) $\Rightarrow$ dislocations are not thermodynamically stable (cannot be removed by cooling) but are **non-equilibrium** defects.

- **Peierls–Nabarro stress** (lattice friction) : $\tau_p = [2G/(1-\nu)]\, \exp[\,-2\pi\, w/b\,]$, $w = a/(1-\nu)$ = width of the dislocation core. Wide core (FCC metals) $\Rightarrow$ small τ_p ; narrow core (covalent, BCC screw) $\Rightarrow$ large τ_p.

7. Forces on and between Dislocations

Force on a dislocation (Peach–Koehler)

An applied shear stress τ on the slip plane in the direction of b exerts a force per unit length of line, perpendicular to the line :

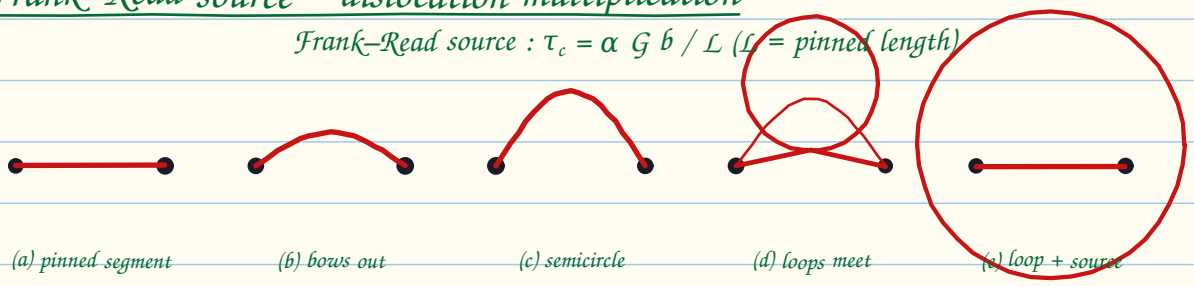
$$F = \tau\, b \text{ (per unit length) ; vector form } \mathbf{F} = (\boldsymbol{\sigma} \cdot \mathbf{b}) \times \boldsymbol{\xi}$$

- Derivation (work) : the slip of the area $\ell \cdot dx$ gives displacement b over area $\ell\, dx$ $\Rightarrow$ work = $\tau(\ell\, dx)\, b = F\, dx \Rightarrow F = \tau\, b\, \ell$. Direction – always normal to the dislocation line, toward the unslipped region.
- **Bowing** : a segment of length L pinned at both ends bows to a radius R under τ : $\tau\, b = T/R \Rightarrow R = Gb/(2\tau)$ (using $T = \tfrac{1}{2}Gb^2$). Minimum radius $L/2 \Rightarrow$ critical stress to operate a source :

$$\tau_c = G\,b / L$$

Frank–Read source – dislocation multiplication

Frank–Read source : $\tau_c = \alpha\, G\, b / L$ (L = pinned length)



- A segment between two pinning points (nodes, precipitates) bows out under stress (a), (b) ; at the semicircle (c) stress is maximum ($R = L/2$, $\tau_c = Gb/L$) ; beyond it the loop expands unstably, parts meet (d) and annihilate to form a complete loop and regenerate the original segment (e) – one source emits many loops ⇒ rapid increase of ρ ($10^6 \rightarrow 10^{12}$ cm^{-2}) during plastic deformation and cold working.

Ex 5. Cu, $L = 1\ \mu m$: $\tau_c = (48\times10^{9})(2.55\times10^{-10})/10^{-6} = 12.2$ MPa ; for $L = 10\ \mu m$: 1.2 MPa. (Shows shorter segments need higher stress – source hardening.)

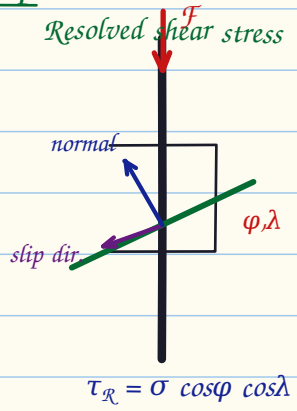
Interaction between parallel dislocations

- **Parallel screws** (b_1, b_2, distance r) : $F = G\, b_1 b_2/(2\pi\, r)$ per unit length – repulsive for like sign, attractive for opposite.
- **Parallel edges on the same glide plane** (separation x) : $F = G\, b_1 b_2/(2\pi(1-\nu)\, x)$ – like signs repel, unlike signs attract and annihilate when they meet (vanish if on the same plane).
- **Parallel edges on different planes** (offset y) : $F_x = [G\, b_1 b_2/2\pi(1-\nu)] \cdot x(x^2 - y^2)/(x^2 + y^2)^2$. Like edges are stable (a stable low-energy configuration, **tilt/small-angle boundary**) at $x = 0$ – one directly above the other ($\perp$ above $\perp$: polygonisation).
- Consequences : (i) **pile-up** at a barrier (grain boundary) – n dislocations with stress concentration $\tau' = n\tau$ ($n = \pi\tau\, L/(G\, b)$ ...) leads to slip transmission (Hall–Petch, Unit II) or crack nucleation ; (ii) interactions with forest dislocations ⇒ work hardening $\tau = \tau_0 + \alpha\, G\, b\, \sqrt{\rho}$ ($\alpha \approx 0.2$–0.5) ; (iii) dislocation–solute (Cottrell atmospheres), dislocation–precipitate interactions ⇒ strengthening.

Ex 6. Cu, $\rho = 10^{15}\ m^{-2}$ (cold-worked) : Taylor $\Delta\tau = 0.5 \times 48\times10^{9} \times 2.55\times10^{-10} \times \sqrt{(10^{15})} = 0.5 \times 12.24 \times 3.16\times10^{7} =$ 193 MPa.

8. Slip & Twinning

Slip



Schmid's law : slip starts when $\tau_R = \tau_{CRSS}$

Schmid factor $m = \cos\varphi \cos\lambda \le 0.5$ ($\varphi=\lambda=45°$)

$\sigma_y = \tau_{CRSS}/m$ (single crystal)

CRSS (MPa) : Cu 0.6–1 ; Al 0.8 ; Zn 0.18 ; Fe ≈ 30 ; Mg 0.8

slip planes : most densely packed ; directions : closest packed

multiple slip → work hardening (stage II)

- **Slip** = shear of one part of the crystal over another along a crystallographic **slip plane** in a **slip direction** by the glide of dislocations. Visible as slip lines/bands (each step ≈ 100–1000 atomic spacings). Slip distance is an integral multiple of the lattice repeat.
- **Schmid's law** : slip begins when the resolved shear stress on the slip system reaches a critical value (CRSS) — a material constant at a given T :

$$\tau_R = (P/A)\cos\varphi\cos\lambda = \sigma\cos\varphi\cos\lambda \quad \sigma_y = \tau_{CRSS}/(\cos\varphi\cos\lambda)$$

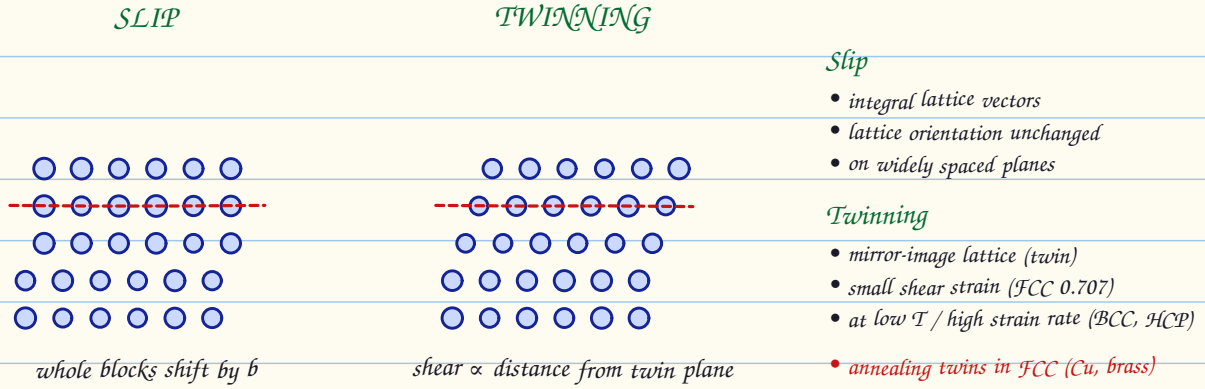
where φ = angle between the tensile axis and the slip-plane normal, λ = angle between the tensile axis and the slip direction ($\varphi + \lambda$ need not be 90°). **Schmid factor** $m = \cos\varphi\cos\lambda \le 0.5$.

- The system with the highest m slips first (primary system). Ex : Zn single crystal yields at very low σ when $m \approx 0.5$, but is hard when the basal plane is parallel ($\varphi = 0°$) or perpendicular ($\lambda = 90°$) to the axis. In polycrystals the average $m \approx 0.31$ (Taylor factor ≈ 3.06 for FCC).

Ex 7. Cu single crystal : $\varphi = 45°$, $\lambda = 35°$, applied $\sigma = 2$ MPa : $m = \cos45° \cos35° = 0.7071 \times 0.8192 = 0.579 \Rightarrow \tau_R = 2 \times 0.579 =$ 1.16 MPa (> CRSS 0.6–1 MPa ⇒ slips). The tensile yield stress for this orientation = 0.98/0.579 = 1.7 MPa.

- Effects of slip : crystal rotation toward the tensile axis (slip direction aligns with load), lattice curvature, work hardening (stage I easy glide, II linear hardening, III dynamic recovery by cross-slip). **Slip bands** may be coarse (wavy vs planar slip depending on γ_{SF}).
- **Kink bands** : localised bending (hcp crystals loaded parallel to basal plane).

Twinning



- **Twinning** = a homogeneous shear of a region of the crystal such that the deformed region is the mirror image of the parent across the **twin plane** ; every atomic plane moves by an amount ∝ its distance from the twin plane (fraction of a lattice vector) – unlike slip.
- Twin systems : FCC {111}⟨11-2⟩ (shear $s = 1/\sqrt{2} = 0.707$; by passage of a/6⟨112⟩ Shockley partials on successive planes) ; BCC {112}⟨111⟩ ($s = 0.707$) ; HCP {10-12}⟨10-11⟩ (Zn, Mg, Ti).

Feature	Slip	Twinning
Lattice orientation	unchanged (steps)	reoriented (mirror image)
Shift	integral lattice vectors	fraction of lattice vector
Occurs on	widely spaced planes	every plane in the twinned region
Strain produced	large	small (< 10 %)
Conditions	any temperature	low T, high strain rate, few slip systems (BCC, HCP) ; FCC low γ
Rate	moderate	very fast, audible "tin cry"
Tool	dislocation glide	twinning dislocation / pole mechanism

- Types : **mechanical (deformation) twins** (BCC Neumann bands in Fe, HCP) and **annealing twins** (FCC Cu, brass – formed during recrystallisation, straight parallel-sided). Roles : reorient crystals so that further slip becomes possible ; accommodate strain in HCP ; trigger cleavage cracks in BCC at low T. Twinning stress is relatively temperature independent (stress to nucleate high, to propagate low).

9. Additional Solved Problems

P1. A tensile test on a steel bar of 12 mm dia and gauge length 50 mm : load at yield 56.5 kN, fracture dia 7 mm, maximum load 75 kN. $A_0 = 113.1\ mm^2 \Rightarrow \sigma_y = 56500/113.1 = 500\ MPa$; $UTS = 75000/113.1 = 663\ MPa$; $\%\ RA = (113.1 - 38.5)/113.1 \times 100 = 66\ \%$.

P2. True strain from engineering strain 0.20 : $\varepsilon_T = \ln 1.2 = 0.182$; $\sigma = 300\ MPa \Rightarrow \sigma_T = 360\ MPa$.

P3. Burgers vector of Al ($a = 4.05$ Å) : $|b| = a/\sqrt{2} = 2.86$ Å ; Mo ($a = 3.15$ Å, BCC) : $|b| = 3.15 \times 0.866 = 2.73$ Å.

P4. Force per unit length on a dislocation (Cu, $b = 2.55$ Å) under $\tau = 10\ MPa$: $F = \tau\, b = 10\times10^{6} \times 2.55\times10^{-10} = 2.55\times10^{-3}\ N/m$.

P5. Dislocation density $10^{10}\ m^{-2} \Rightarrow$ spacing $\approx 1/\sqrt{\rho} = 10\ \mu m$; at $10^{15}\ m^{-2} \Rightarrow 0.03\ \mu m$ (30 nm).

Short answers – Quick Revision

- *Why do metals yield far below the theoretical shear strength?* Plastic flow occurs by glide of dislocations, breaking only a few bonds at a time.
- *Is E changed by cold working?* No (structure-insensitive) ; yield and tensile strength are structure-sensitive.
- *Why can screw dislocations cross-slip but edge cannot?* $b \parallel \xi$ means no unique glide plane for the screw.
- *Why is FCC more ductile than HCP?* 12 equivalent slip systems versus only 3 basal.
- *Why are dislocation energies proportional to b^2?* Elastic strain energy $\propto$ (strain)^2, strain $\propto b$.

10. Quick Formula Sheet – Unit I

Item	Formula
Hooke's law	$\sigma = E\varepsilon$; $\tau = G\gamma$; $E = 2G(1+\nu) = 3K(1-2\nu)$
True stress/strain	$\sigma_T = \sigma(1+\varepsilon)$; $\varepsilon_T = \ln(1+\varepsilon)$
Theoretical strength	$\tau_{th} \approx G/2\pi$
Screw stress	$\tau_{\theta z} = Gb/2\pi r$
Energy per length	$E \approx \frac{1}{2}Gb^2$; screw $Gb^2/4\pi \cdot \ln(R/r_0)$
Force on dislocation	$F = \tau b$
Frank–Read	$\tau_c = Gb/L$
Taylor hardening	$\Delta\tau = \alpha G b \sqrt{\rho}$
Schmid's law	$\tau_R = \sigma \cos\varphi \cos\lambda$
Partial separation	$d = Gb_1 \cdot b_2/(2\pi\gamma)$

11. Important Questions

- State Hooke's law; derive the relation between E, G, K and ν. Explain the origin of elastic behaviour from the interatomic force curve.
- Why is the observed yield strength much lower than the theoretical? Explain the need for dislocations.
- Differentiate edge and screw dislocations ; define Burgers vector and Burgers circuit ; explain dislocation loops.
- Describe dislocations in FCC, BCC and HCP lattices; Burgers vectors; partial dislocations & stacking fault.
- Derive the stress field and energy of a screw dislocation; explain forces between dislocations; Frank–Read source.
- State Schmid's law with derivation. Compare slip and twinning.